$$\begin{aligned}\tan\left(\frac{\pi}{4}-\frac{x}{2}\right) &= \frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}} \\ &= \frac{1-\sin x}{\cos x} \\ &= \frac{1}{\cos x}-\frac{\sin x}{\cos x} \\ &= \sec x - \tan x\end{aligned}$$

(i) Using $(*)$ with $x = \dfrac{\pi}{4}$,

$$\tan\frac{\pi}{8} = \sec\frac{\pi}{4} - \tan\frac{\pi}{4} = \sqrt{2}-1$$

$$\begin{aligned}\tan\frac{11\pi}{24} &= \tan\left(\frac{\pi}{3}+\frac{\pi}{8}\right) \\ &= \frac{\sqrt{3}+\sqrt{2}-1}{1-\sqrt{3}(\sqrt{2}-1)} \\ &= \frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1}\end{aligned}$$

(ii)

$$(\sqrt{3}-\sqrt{6}+1)(2+\sqrt{2}+\sqrt{3}+\sqrt{6}) = \sqrt{3}+\sqrt{2}-1$$

$$\therefore \frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1} = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}$$

(iii)

Using $(*)$ with $x = \dfrac{11\pi}{24}$,

$$\tan\frac{\pi}{48} = \sec\frac{11\pi}{24} - \tan\frac{11\pi}{24}$$

$$\begin{aligned}\sec\frac{11\pi}{24} &= \sqrt{1+\tan^2\frac{11\pi}{24}} \\ &= \sqrt{1+(2+\sqrt{2}+\sqrt{3}+\sqrt{6})^2} \\ &= \sqrt{16+10\sqrt{2}+8\sqrt{3}+6\sqrt{6}}\end{aligned}$$

From part **(ii)**, $\tan\dfrac{11\pi}{24} = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}$

$$\therefore \tan\frac{\pi}{48} = \sqrt{16+10\sqrt{2}+8\sqrt{3}+6\sqrt{6}} - 2 - \sqrt{2} - \sqrt{3} - \sqrt{6}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: July 2, 2026